



FACULTY OF ACCOUNTING AND INFORMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY

Bachelor in ICT

Study Guide

2026

Research Skills

RESK301
SAQA CREDITS: 1.04

Name of Lecturer : Miss Lindelweyizizwe Manqele
Office : DBO207S, West Wing, Block B
Campus location : Ritson
E-Mail : Lindelweyizizwem@dut.ac.za
Consultation times with Lecturer: email to book a consultation slot/book via Class rep

Head of Department : N/A
Programme coordinator : Mrs FT Khan
Campus location : Ritson
Departmental Secretary : Ms T Ntuli
Contact details : itdept@dut.ac.za
thabilen1@dut.ac.za
031 373 5596/5594

Lectures : 1 and 2 double periods per week
Consultation : 1 compulsory fortnightly consultation with the lecturer

Duration of lecture : 50 minutes each
Consultation hours : Thursday 13:00-16:00 and Friday 10:00-12:00

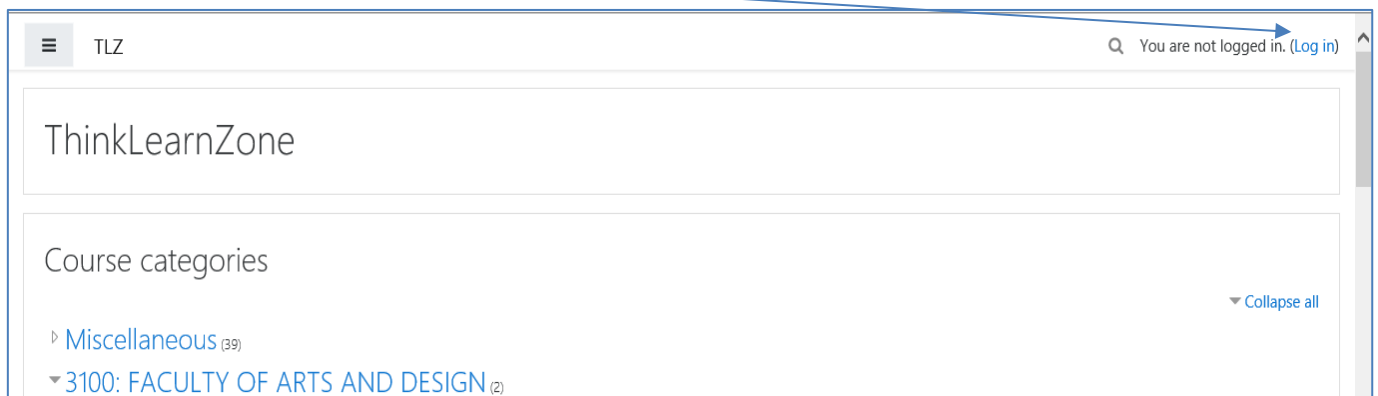
1. Welcome

This study guide provides important information regarding the requirements, administration, and expectations for this module.

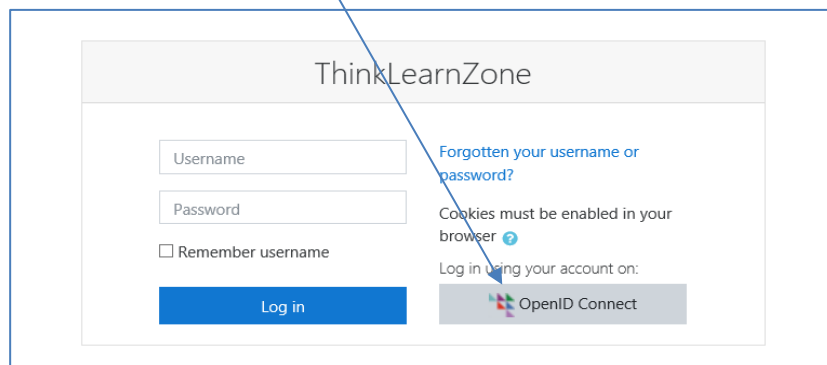
2. Using your online ThinkLearnZone portal

All taught subjects/modules have their own online portal on the ThinkLearnZone. You can access your portal at <https://tlzprod.dut.ac.za>

Click on (Log in) on top right hand corner



On the next screen click on “OpenID Connect”



Use your dut4life email address to sign (yourstudentnumber@dut4life.ac.za) and your email password as your password (Dut + first six digits of your ID no., e.g. Dut980610. For first-time 2020 students - use \$\$Dut + first six digits of your ID no., e.g. \$\$Dut980610)

Ask your lecturer for guidance or check out “how to log in” on the DUT e-learning website https://elearning.dut.ac.za/faq/faq_students/

The e-learning website also has contact information for help and technical assistance <http://elearning.dut.ac.za/contacts/>

You can call the e-learning helpdesk on 031 373 6758 or email them on tlzsupport@dut.ac.za

3. Virtual Help Desk

A **Virtual Help Desk** will be set up to provide students with **academic and administrative support** in a more accessible and efficient manner.

The Virtual Help Desk is intended to assist students who may be experiencing challenges or require guidance without the need to visit campus physically.

Purpose of the Virtual Help Desk

The Virtual Help Desk aims to:

- Provide **quick and reliable support** to students
- Reduce delays in resolving student queries
- Improve communication between students and the Department
- Support students with both **academic and administrative matters**

Types of Support Available

Students will be able to receive assistance with:

- Registration and academic administration queries
- Module and timetable-related issues
- Assessment and examination enquiries
- General academic guidance and referrals
- ICT-related queries (where applicable)

Please note: Queries outside the scope of the Help Desk will be referred to the relevant office or staff member.

How the Virtual Help Desk Will Operate

- **Mode:** Virtual
- **Platform:** MS Teams Chat, MS Teams Call Service, Email Support
- **Operating Hours:**
Monday to Friday: (08:00 – 22:00)
- **Start Date:** 19.02.2026

How to Access the Virtual Help Desk

Students will be able to:

- Submit queries through your DUT student email address / gmail or yahoo email address.

All queries will be attended to on a **first-come, first-served basis**.

General Enquiries

Email: itdept@dut.ac.za

Primary Contact Persons

- Sthembiso Mthethwa – 22050584@dut4life.ac.za
- Thandokuhle Mbokazi – 22111885@dut4life.ac.za
- Nosipho Mngoma – 22263084@dut4life.ac.za
- Asande Msiya – 22205216@dut4life.ac.za
- Nokwazi Gumede – 22211795@dut4life.ac.za
- Ntandoyenkosi Ntshangase – 22208552@dut4life.ac.za
- Liso Blaai – 22283725@dut4life.ac.za

Secondary Contact Persons

- Bhuty – bhutym@dut.ac.za | Tel: 031-373-5252
- Geeta – geetap@dut.ac.za | Tel: 031-373-5560

Qualification-Specific Enquiries

- Diploma DIIAD1 – Dr. Jay Pancham – panchamj@dut.ac.za
- Advanced Diploma ADICTI – Kesarie Singh – kesaries@dut.ac.za
- Diploma (4 years) DIIAFI – Ghulam Mohamed – ghulamM1@dut.ac.za
- Degree BINCT1 – Fatima Thobeka Khan – mqadi@dut.ac.za
- Honours BICTH1 – Lindelweyizizwe Mangele – lindelweyizizwem@dut.ac.za
- Masters and PhD – Robyn Thompson – robynt@dut.ac.za

Important Guidelines

- Ensure your **student number and full name** are included in all communications
- Be clear and concise when submitting queries
- Allow reasonable time for responses, especially during peak periods
- Maintain respectful and professional communication at all times

Further Communication

Additional details, including the **official launch date and access instructions**, will be communicated to students via:

- Student email
- Departmental notices
- Learning Management System (LMS)

The Department encourages all students to use the Virtual Help Desk as a support resource to enhance their academic journey. Your cooperation and engagement will help ensure the success of this initiative.

Issued by:

Department of Information Technology

Durban University of Technology

Date: 19.02.2026

4. Introduction to the module

Research Skills aims to advance interpretive skills, methodical application of techniques, and adherence to the principles and philosophies of IT research. This module is designed to facilitate the student's development of skills in conducting high-quality research.

Module content is continuously updated to align with contemporary knowledge and developments in IT research. As the module progresses, students will be required to demonstrate skills in identifying a persistent problem or gap evident in an IT research area, devising a solution, and writing about the research undertaking. More specifically, the student should be able to formulate a research question or hypothesis, aim and objectives, conduct a critical review of literature, apply relevant data collection and analysis techniques, and evaluate the research.

1. Learning outcomes

Upon completion of this module, the student will be equipped with Knowledge of research concepts and will practice applying them to contribute to IT discourse:

LO1: Identifying a persistent problem or gap evident in an IT research area (**problem identification**)

LO2: Review existing literature, utilizing resources available (**Reading skills**)

LO3: Learning formal **writing skills** by writing research output while taking into consideration ethics

LO4: Skill of **selecting suitable research methodology** and research methods to address the problem identified.

This module facilitates its learners' exploration of recent and emerging methodologies, practices, and paradigms in IT research. See Appendix A for learning outcomes and assessment criteria. LO1 (Learning outcome 1)

2. Learning, teaching, and assessment strategies

a) Learning activities

Learning activities include contact lectures, tutorials, assignments, self-learning, and online assessments and tutorials (terms and conditions apply to online submissions). The assignment requires the professional presentation of a developed research artefact. Submission of assessments and tutorials via online tools. Seminars, presentations, and workshops from staff at DUT Library on research tools and databases.

b) Graduate attributes

This course strives to develop the following graduate attributes:

- Active and reflective learners through engagement with recent and relevant developments in IT-based research methods and emerging topical areas.
- Critical and creative thinkers who can independently apply their developed knowledge in research methods and collaborate with participants and other researchers.
- Effective communicators of their rationale and defense of the choice of research methods
- Practiced research and scientific writing skills

c) Assessment

The following sections highlight the assessment strategies.

Tutorials

Students are provided with tutorials for each section of the curriculum. Students are strongly advised to attempt all tutorials. Although memos will not be made available, responses to these tutorials may be shown to the lecturer or tutor for commentary.

Test and make-up test

Four tasks are included as part of the assessment. Students who do not write a test/task must provide valid proof of absenteeism (e.g., medical certificate, proof of incarceration) within 5 working days of the absenteeism to score the full mark obtained in the make-up test. A student who failed a test may attempt the make-up test. However, their mark will be *capped at 50%*. Please note that only one make-up test is scheduled. If a student misses both tests, one of the test marks will remain zero.

All tasks will contribute towards the module course mark. Students must identify an IT research problem or opportunity, formulate a research problem statement, state the aim, prepare a short overview of relevant literature, review the relevant literature, and create a research poster.

Final mark

The final mark will be calculated as follows:

Course mark (DP)	Objectives	Deliverables	%
Assessment 1	Problem Identification	Problem Statement	25
Assessment 2	Review existing methods in the literature	Systematic literature review	25
Assessment 3	Review Articles (Spot Test)	A comprehensive review	25

Assessment 4	Select a solution suitable to address the problem identified	Solution Selection Method	25
--------------	--	---------------------------	----

**Subject to change*

d) Activities to promote learning

Lecture attendance is vital for developing a core understanding of Research Skills, concepts, and assessment requirements. In-class case studies, tutorials, and discussions enhance the learning experience.

Students are required to access the online learning platform, which will disseminate important information and course material.

- i. Students may engage in online discussions about course content, assignments, test preparation, and tutorials. It is common for students to form social media chat groups. This is a powerful medium for peer-to-peer collaboration and assistance.
- ii. Tutorials help students develop knowledge pertinent to assessments and topical areas. Students are advised to attempt all tutorials.
- iii. Research skills sessions are more of a supervision session; students are encouraged to attend **ALL** the sessions. The attendance mark will apply to each assessment.
- iv. **NO LATE SUBMISSION** will be considered. Students must start working on their projects on time.
- v. There would be marks allocated for **progress review** (supervision) per group

e) Use of the library

The library offers access to textbooks, online journals, and other relevant publications. Tools such as EndNote and Turnitin are also available.

5. Scheme of work

09 Feb	LESSON 1	Introduction to research & discussion on the student guide
12 Feb		Stages of research
13 Feb		Information sources and retrieval & Library demonstration
16 Feb		Academic referencing & article search demonstration
19 Feb		Problem Identification
20 Feb		Assessment 1 project guide: TASK 1
11 Mar	LESSON 2	SLR – Keywords and AI Tools
12 Mar		SLR- Screening and analysis
19 Mar		Analysis and synthesis

20 Mar		Results and discussion
26 March		Summary of articles, Conclusion, Acknowledgement and Appendices
		Vacation
10 April		Assessment 2 submission: TASK 2
16 April	LESSON 3	Article review and call reading
19 April		Article review
		Assessment 3 – Spot test: TASK 3
April	LESSON 4	Solution Selection
1 May		Assessment 4- Research Methodology: TASK 4
7 May		Exam preparation
8 May		Exam preparation
15 May		DP submission
16 May		Study period
		Examination

1. Copyright and plagiarism

Plagiarism is the use of any sources of information, including but limited to books, journal articles, extracts from thesis, online documentation, website material, or any other material written by someone else and presented as the student's own. Plagiarism includes the 'copying and pasting' of information from any source which is then presented as the student's own. Plagiarism also includes copying of other students' assignments, group assignments, projects and any assessments and presenting this as the student's own.

plagiarism will result in a severe penalty in accordance with the DUT's Plagiarism Policy

In escalated circumstances, plagiarism may lead to an institutional disciplinary hearing and a fine imposed on the student.

A plagiarism declaration will be made. This declaration must be completed and signed by each student and attached to every assignment handed in for assessment. Assignments will not be marked if this completed plagiarism declaration is not attached.

2. Student support

Students are advised to set appointments for lecturer consultations. Difficulties with any aspect of the syllabus or the module may be addressed during consultation times.

Students that are performing poorly will be called in for special meetings with the lecturer to derive means of improving their performance in the module and discussing any experienced difficulties.

3. Work Integrated Learning (WIL), industry, community, and occupation-related information.

There is no WIL component for this module.

4. Quality assurance and enhancement

Students will have the opportunity to provide feedback on the module's quality and their learning experience via the PRE001/002 lecturer and module evaluations, as requested by the CPQA. Students are invited to extend comments, criticisms, or constructive contributions to the lecturer about module delivery and content. In addition, there will be student representative meetings once per term.

5. Useful resources

There are no prescribed texts for this module. The following list of books is helpful and available in the MLS library:

1. Creswell, J. W. 2018. Research design: qualitative, quantitative, and mixed methods approaches Fifth edition. Sage.
2. Dane, F.C. 2018. Evaluating research: methodology for people who need to read research Second edition. Sage.
3. Myers, G. Research methodology: the beginner's manual. 2017. Acacia Private Publishing.
4. Kumar, R. 2019. Research methodology: a step-by-step guide for beginners Fifth edition. Sage.
5. Locharoenrat, K. 2017. Research methodologies for beginners. Pan Stanford Publishing.

The lecturer will point out other relevant publications. The student is responsible for locating and downloading a copy, which may form part of the examinable material.

6. Learning outcomes and assessment criteria

Learning outcomes	Assessment criteria
1. Identify the characteristics and components of academic research. 2. Evaluate research activities concerning ethical/unethical conduct. 3. Identify IT research problems or opportunities. 4. Identifying different sources of information 5. Evaluate the reliability of different sources of information 6. Apply referencing styles in the acknowledgment of sources. 7. Article reviewal 8. Evaluate different research strategies. 9. Evaluate and elect a suitable research dataset. AND/OR 10. Evaluate different data collection methods. 11. Evaluation criteria 12. Compile the research output: Poster	1. Demonstrate knowledge of the typical stages and the process of contemporary IT research. 2. Identify and take corrective action for unethical research activities. 3. Formulate a research question, aim, and objective for a research undertaking. 4. Synthesize information from multiple sources. 5. Differentiate between reliable and unreliable sources of information. 6. Demonstrate the consistent application of a referencing format. 7. Demonstrate the skill of reviewing research articles 8. Demonstrate the correct solution selection skill for a research undertaking. 9. Identify relevant datasets to address the problem at hand AND/OR 10. Demonstrate the ability to ensure the reliability and validity of data collection instruments and data that need ethical clearance. 11. Apply relevant metrics for testing, computational mathematics, coding, and analytic induction + grounded theory to analyze data. 12. Demonstrate the research practice